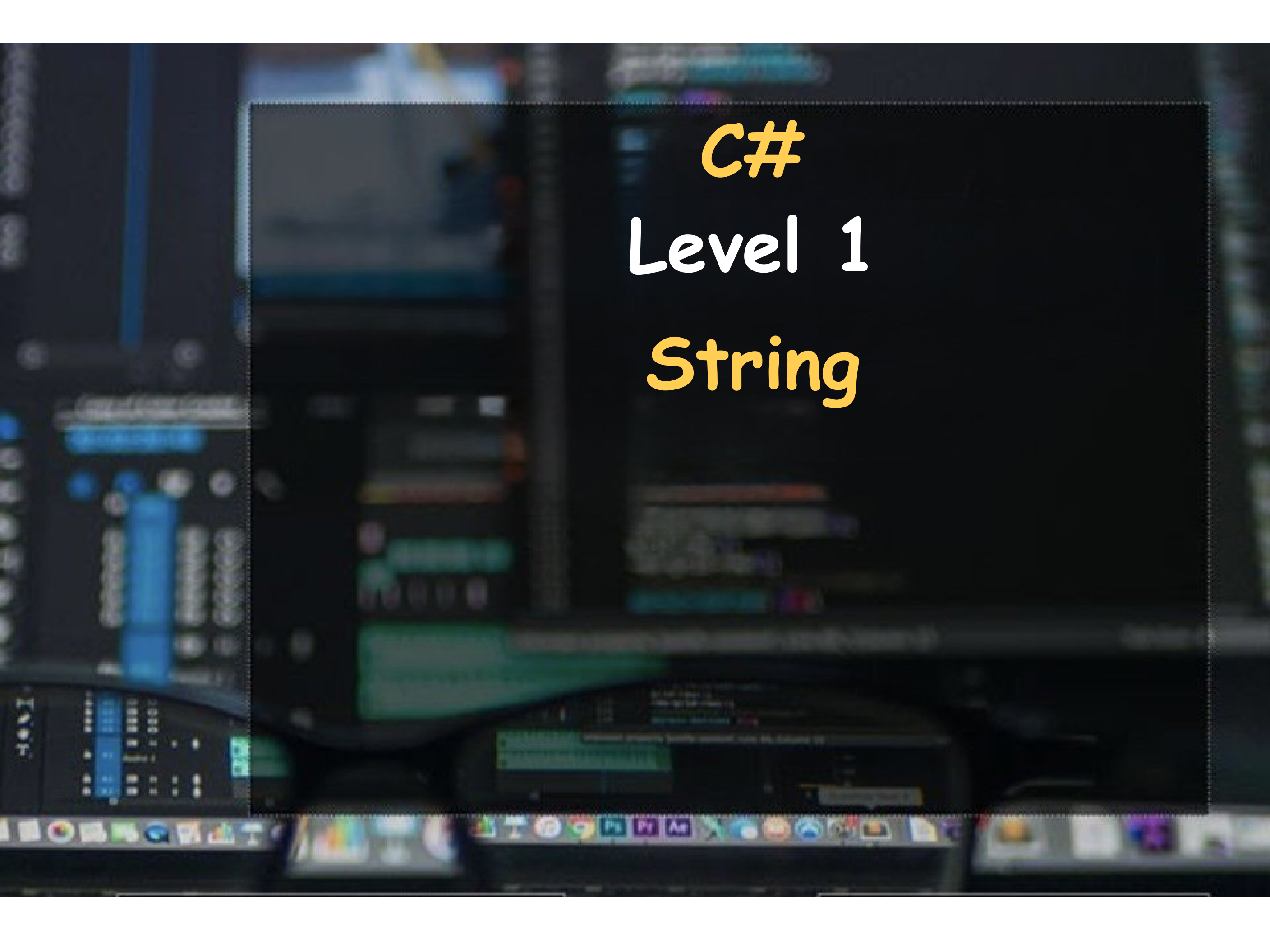


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, yellow). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, with the text overlaid on top.

C# Level 1 String



C#

Level 1

String

Definition

String

-  ``string`` is used to store a sequence of characters.

-  C# provides many methods for searching, modifying, and extracting parts of a string.

Explanation

- Length `.Length` returns Total characters (including spaces).
- Substring `.Substring(startIndex, length)` Extracts part of the string.characters.
- Case Conversion
 - `.ToLower()` converts the string to lowercase.
 - `.ToUpper()` converts the string to uppercase.
- Indexer `S1[index]` Accesses character at index (starts at `0`).
- Insert `.Insert(index, value)` Inserts text without removing existing characters.
- Replace `.Replace(oldValue, newValue)` Replaces all occurrences.
- Searching
 - `.IndexOf(value)` returns the index of the first occurrence.
 - `.LastIndexOf(value)` returns the index of the last occurrence.
 - `.Contains(value)` Returns `true` if found, else `false` .
- Split `.Split(delimiter)` Splits string into an array.
 - The resulting elements can be accessed using their indexes.
- Trim
 - `.Trim()` removes whitespace from both the beginning and end.
 - `.TrimStart()` removes whitespace from the beginning.
 - `.TrimEnd()` removes whitespace from the end.

Procedures

1. Create a string.
2. Use string methods to get its length, extract text, change case, search, or modify it.
3. Use `.Split()` when the string contains multiple values separated by a delimiter.
4. Use `.Trim()` methods to remove unwanted whitespace.

💡 Example

```
string S1 = "Sara Alterkawi";  
S1.Length → `14`  
S1.Substring(2, 5) → `ra Al`  
S1.ToLower() → `sara alterkawi`  
S1.ToUpper() → `SARA ALTERKAWI`  
S1[2] → `r`  
S1.Insert(3, "KKKK") → `SarKKKKa Alterkawi`  
S1.Replace("a", "*") → `S*r* Alterkawi`  
S1.IndexOf("r") → `2`  
S1.Contains("t") → `True`  
S1.Contains("x") → `False`  
S1.LastIndexOf("a") → `12`  
string S2 = "Ahmad, Hadi, Aljaseem";  
S2.Split(',') →  
`NamesList[0]` → `Ahmad`  
`NamesList[1]` → `Hadi`  
`NamesList[2]` → `Aljaseem`  
string S3 = "  Alterkawi  ";  
S3.Trim() → `Alterkawi`  
➡ Result
```

The methods allow you to extract, search, modify, split, and clean string values.

Note

- ◆ String indexes start from `0`.
- ◆ Spaces are counted as characters by `.Length`.
- ◆ `.Split()` returns an array of strings.
- ◆ `.Contains()` returns a `bool`.

⊘ Warning

- ◆ `Substring()` throws an exception if the starting index or requested length is outside the string range.
- ◆ `IndexOf()` returns `-1` when the searched value is not found.
- ◆ `.Replace()` replaces all matching occurrences, not just the first one.
- ◆ `.Split(',')` does not automatically remove spaces after the comma, so `" Hadi"` contains a leading space. Use `.Trim()` if needed.

✦ Summary

- ✓ ``string`` stores text and provides methods for working with it.
- ✓ Searching uses ``IndexOf()``, ``LastIndexOf()``, and ``Contains()``.
- ✓ Text can be modified using ``Insert()`` and ``Replace()``.
- ✓ ``Split()`` converts separated text into a string array.
- ✓ ``Trim()`` methods remove unwanted whitespace.

Thank you 😊